

Meeting 0107

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12/17 Discussion

Semiconductor Manufacturing Process

► Unbox black-box function $\hat{f}$

- 在 PCA 平面（有拿掉極端值），clusters 內的 0/1 有可分性
- 選定「有故事的」 X_0 作為敘事地標（landmark）
- 上次有用 X_{test} ，下次可選擇 $\hat{p}(x|x \in X_0) \approx 0.5$ 的點

► Decision strategy

- fail 來自 FAR 低的 cluster $\Rightarrow$ 高可信度的 fail
- fail 來自 FAR 高的 cluster $\Rightarrow$ 需優先調查該 cluster

12/17 Discussion

► 討論 local R^2 與 $\hat{p}(x_0)$ 的關係

- local R^2 衡量的是 local surrogate $\hat{g}$ 對黑箱 $\hat{f}$ 的局部擬合度
- 因此 $\hat{p}(x_0) \approx 0.5$ 時的 local R^2 很高並不矛盾；若在 $\hat{p}(x_0) \rightarrow 0/1$ （黑箱更確定時），local R^2 下降，表示目前的局部模型 $\hat{g}$ 在那些區域不足以逼近 $\hat{f}$ ，需要更強的 $\hat{g}$

12/17 Todo

► Separate and fit black-box function

- 將資料切分： $\mathcal{D}_c = \{(x_i, y_i) : c_i = c\}$ ，其中 $c \in \{1, \dots, K\}$ ，
對每個 cluster 獨立訓練 black-box $\hat{f}_c : x \mapsto \hat{p}_c(x)$

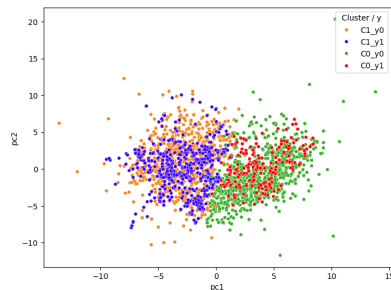
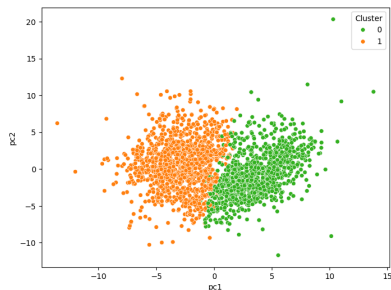
► Choose other x_0

- 選擇 $\hat{p}(x_0) \approx 0.5$ 的點，作為 local 的解釋地標
- 對 x_0 做 local surrogate (生成 $x_\ell = x_0 + \varepsilon_\ell$)，計算
 $y_\ell = \hat{f}_c(x_\ell)$ ，最後 fit 模型 $\hat{g}$

► Try other ML as $\hat{g}$

- 使用 Logistic 作為 black-box，XGBoost 作為 $\hat{g}$

Results I SMOTE + PCA + KMeans

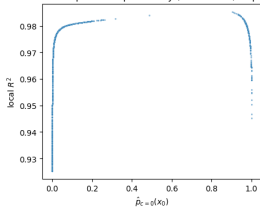


cluster	0	1
$y = 0$	743	687
$y = 1$	205	510

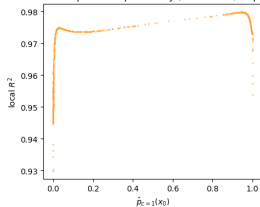
Separate and build local surrogate models

- ▶ 以（標準化後、經 SMOTE 平衡的）資料 PCA+KMeans 將樣本分群，得到 $c \in \{0, 1\}$ ；在各群內分別訓練 black-box $\hat{f}_c$
- ▶ 給定屬於群 c 的設計點 x_0
- ▶ 生成局部擾動樣本： $x_\ell = x_0 + \varepsilon_\ell$ ，其中 $\varepsilon_\ell \sim N(0, \sigma^2 I)$
- ▶ 代入對應的 cluster black-box： $y_\ell = \hat{f}_c(x_\ell)$
- ▶ 以 $\{(x_\ell, y_\ell)\}$ fit local surrogate $\hat{g}(\cdot; x_0)$ ，使得在 x_0 附近內 $\hat{f}_c(x) \approx \hat{g}(x; x_0)$

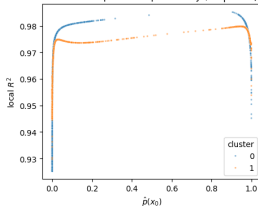
Results II f_{logistic} & g_{linear}

local R^2 vs. predicted probability (cluster = 0, all points)

cluster = 0

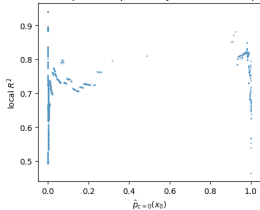
local R^2 vs. predicted probability (cluster = 1, all points)

cluster = 1

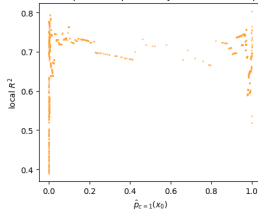
local R^2 vs. predicted probability (all points)

cluster = 0 & 1

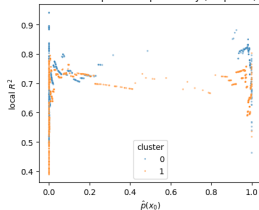
Results III f_{logistic} & g_{decision}

local R^2 vs. predicted probability (cluster = 0, all points)

cluster = 0

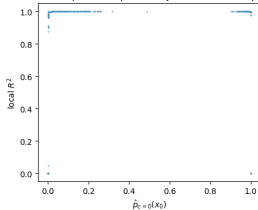
local R^2 vs. predicted probability (cluster = 1, all points)

cluster = 1

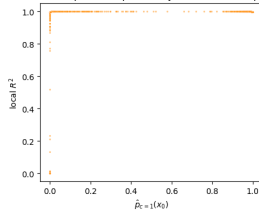
local R^2 vs. predicted probability (all points)

cluster = 0 & 1

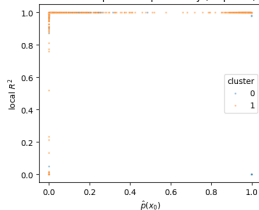
Results IV $f_{\text{logistic}} \& g_{\text{xgb}}$

local R^2 vs. predicted probability (cluster = 0, all points)

cluster = 0

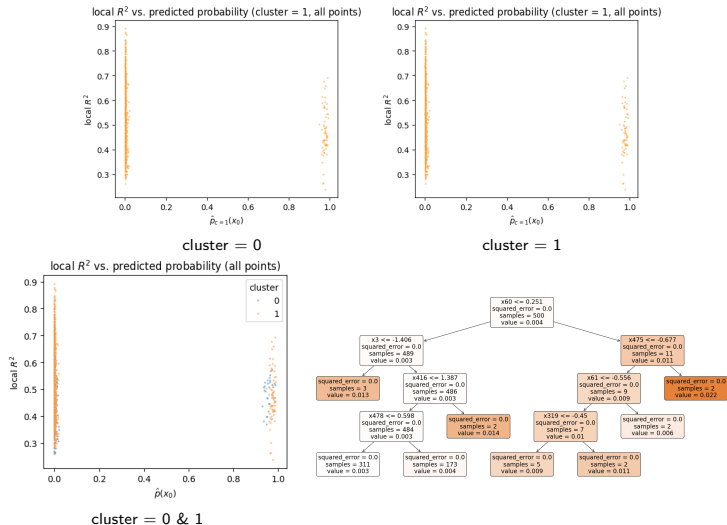
local R^2 vs. predicted probability (cluster = 1, all points)

cluster = 1

local R^2 vs. predicted probability (all points)

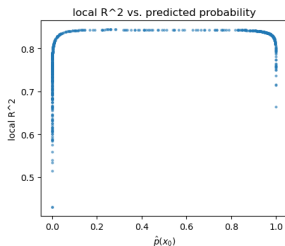
cluster = 0 & 1

Results V f_{xgb} & g_{tree}

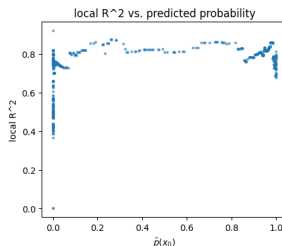


Results VI

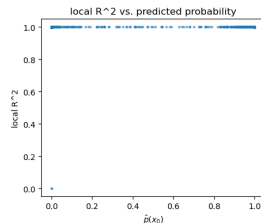
Other ML for $\hat{g}$



$\hat{f}$ is logistic
 $\hat{g}$ is linear



$\hat{f}$ is logistic
 $\hat{g}$ is decision



$\hat{f}$ is logistic
 $\hat{g}$ is XGBoost

The End

Questions & Comments